ECON0054: Economics of Development

Problem Set 2

Suggested Submission Deadline: 14^{th} February 2025

1. From our reading of the Jensen (2010) on education, what instrumental variables does the author use, and why?

   Distance to school in Childhood: According to Card(1995) and Kane&Rouse(1993), students who live farther from schools are less likely to attend and complete more years of education, simply due to geographical and physical challenges and the costs of transportation.

   It also satisfy the exogenous condition. Since the distance to school is not directly related to unobserved factors that influence earnings, except through its effect on education. So, where a student grew up (in terms of proximity to a school) should not directly determine their wages later in life, except through how much schooling they completed.

   Whether received the treatment: It assign a value of 1 to students who received the accurate information of schooling return, and 0 to those who did not receive. It shows relevance because students who received the information were more likely to update their perceived returns, as the study finds a significant increase in their reported expected earnings after the treatment been applied.

   It is also exogenous because treatment assignment was given at random. Since students were randomly chosen to receive information, their assignment is not related to other factors (such as motivation, socioeconomic status, or prior beliefs) that might independently affect their schooling decisions.This means that any change in schooling outcomes due to the IV can be attributed to the effect of updated perceived returns, rather than other unobserved factor.

2. Using your own words, briefly describe what Kremer and Glennester (2011) describe as the "the health as a human capital investment approach"

Kremer and Glennerster (2011) describe the health as a human capital investment approach as the idea that improving health leads to greater economic productivity and long-term benefits for individuals and society.

While Individual will decide on whether to take medicine to prevent certain disease based on their cost-benefit analysis, that is, compare the discounted utility of illness (and time) and the decreased endowment for taking drugs, so not all individuals will choose to prevent, Government should take on interventions as a "investment" in human health, such as vaccinations, deworming, and disease prevention—not just to improve well-being but also as an investment in human capital. Healthier individuals are more likely to attend school, perform better academically, and be more productive in the workforce, ultimately contributing to economic growth.

3. Kremer and Glennester convincingly explain that most governments and donors are generally concerned about the cost of attaining a particular health outcome. Deliver three examples of cost-effective health interventions in poor countries.

Deworming programs costs as little as $0.50 per DALY, significantly improve school participation and long-term earnings(21%-29%). Given the affordability of these programs, still only 10% of infected children are treated among the 400 million.

Vaccination campaigns prevent deadly diseases at a fraction of the cost of treating outbreaks. Measles vaccinations cost about $1 per dose and prevent thousands of deaths each year, making them one of the most cost-effective public health interventions.

Insecticide-Treated Nets (ITNs) have been shown to reduce overall child mortality by up to 38% in malaria-endemic regions (D'Alessandro et al., 1995) and have an estimated cost of less than $50 per DALY saved. Despite their effectiveness, only 19% of children in malaria-affected areas of Africa were protected by ITNs in 2007 (Noor et al., 2009). However, focused policy efforts in recent years have led to a significant increase in ITN coverage rates.

4. Briefly describe Mexico's PROGRESA and its impact on disadvantaged household's health.

Mexico's PROGRESA was launched in 1998 to improve the well-being of disadvantaged households by providing conditional cash transfers. Beneficiary households received financial support equivalent to about one-third of their income, provided they participated in health and nutrition-related activities such as prenatal care, immunization, and nutritional monitoring. The program was initially implemented in rural areas with high poverty levels and was phased in strategically to allow for rigorous evaluation.

PROGRESA significantly improved health outcomes and healthcare utilization among disadvantaged households. Visits to public health clinics in program areas increased by 18.2%, reflecting a shift in health-seeking behavior. Children under the age of three who received support were 22.3% less likely to be reported ill within the previous four weeks compared to those in non-beneficiary households. Longer exposure to the program led to greater cumulative benefits, with children experiencing lower anemia rates and improved growth. Program's success led to its expansion across Mexico, including urban areas, and inspired similar conditional cash transfer programs in nearly 30 other countries. By 2006, it covered five million families, accounting for one-quarter of Mexico's population.

5. Provide an equation for estimating immunisation cost effective interventions.

$$Cost - effective\ Ratio = \frac{Total\ cost\ of\ immunisation\ Programme}{Total\ DALY\ averted}$$

Total cost of immunization program includes vaccine costs, distribution, healthcare personnel, and other implementation expenses. **Total DALYs averted** is the sum of years of life saved and years of disability prevented due to immunization.
A lower cost per DALY averted indicates a more cost-effective intervention, with the WHO considering interventions costing less than GDP per capita per DALY averted as highly cost-effective.

6. According to Banerjee et al (2022), the most effective nudge identified in their study is the combination of three interventions. Which are these?

According to Banerjee et al. (2022), the most effective nudge for increasing immunization rates combines three key interventions:

- **monetary incentives**
- **SMS reminders**
- **seeding ambassadors (information hubs).**

Their study found that this combination is the best, increasing immunization rates by 44% ($p < 0.05$).

To maximize cost-effectiveness, the study suggests that policymakers should opt for l**ow convex incentives, send SMS reminders to 33% of caregivers, and use information hubs to relay messages**. This strategy balances impact and affordability, ensuring the highest number of immunizations per dollar spent.

To maximize the immunization per dollar spent, the best is to use I**nformation hubs with 33% caregivers or more covered of SMS reminder.**

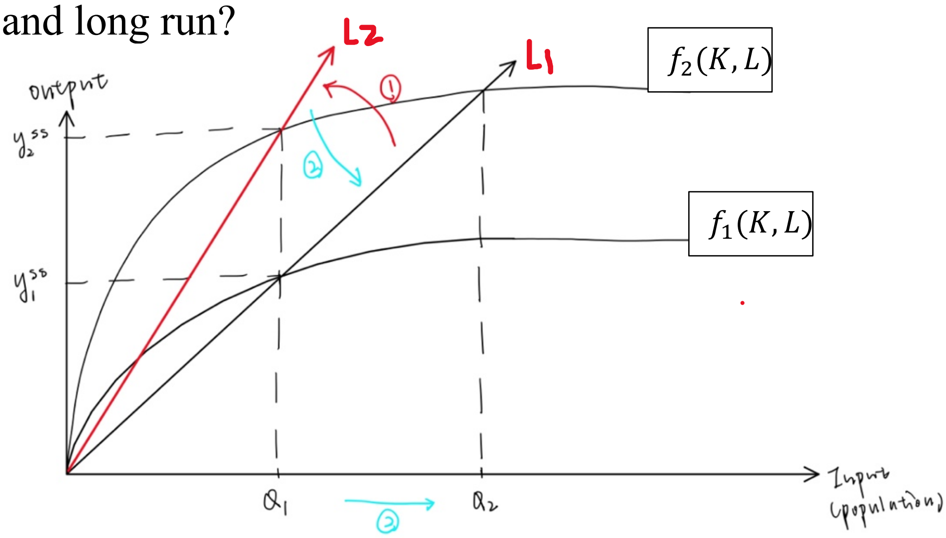
7. In the context of health interventions, what can machine learning be used for?

Disease Prediction: Machine learning can forecasts disease outbreaks by analyzing the past real-time and historical data. By using algorithms to process information from various sources, it can detect early warning signs of potential epidemics. Algorithms have been used to predict influenza outbreaks by analyzing Google search trends and emergency room visits. Similarly, during the COVID-19 pandemic, it helped track infection rates and project future case growth based on mobility data. These allow health authorities to take proactive measures, such as inventing vaccines.

Optimizing Resource Allocation: In many healthcare settings, resources such as vaccines and funding are limited, making efficient allocation critical. It can analyze demographic data, disease prevalence, and healthcare infrastructure to determine where interventions are most cost-effective. So to help optimize resource allocation, that is, money spent on the most cost-effective intervention programs. It can convince policymakers in prioritizing funding for essential health services, ensuring that limited budgets are used effectively to maximize public health impact.

8. Assume that the green revolution in the 1960s doubled the production of agricultural output. What would Malthusians have

predicted on population size and per capita income in the short and long run?



$Cobb - Douglas\ Production\ Function:$

$$F(K,L) = AK^{\alpha}$$

$Given\ that, Green\ Revolution\ in\ 1960s\ doubled\ the\ output,$

$$Y = A_0K^{\alpha} \rightarrow Y = A_1K^{\alpha}$$
$$where\ A_1 = 2A_0$$

$Thus,$

$$Y_2^{ss} = A_1^{\frac{1}{1-\alpha}} \left(\frac{\gamma}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}} = 2A_0^{\frac{1}{1-\alpha}} \left(\frac{\gamma}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}} = 2Y_1^{ss}$$

In **short run**, with no change in population size Q_1, there is a double in Steady State capital income, as a result of more output. However, with the improvements in food availability, the morality rate also increases, and finally there is a population growth, from $Q_1 \rightarrow Q_2$

In **Long run**, with the population increase, the increase of Steady state capital and income are offset by the increasing labor force.
The effect of the Green Revolution on per capita income will be modest, and it may return to levels close to pre-Green Revolution levels

$$K^{ss} = \left(\frac{sA_1}{n_1+\delta}\right)^{\frac{1}{1-\alpha}}$$
$$where \frac{A_1}{n_1} \rightarrow \frac{A_0}{n_0}$$

9. Assume that as a result of massive deportations of illegal migrants, the United States experiences a considerable decline in its working age population. David Weil assumes that the forecast for the period 2010 to 2030 is a decline from 0.60 to 0.54. What will be the annual growth rate of the working age fraction? By how much will GDP per capita per year fall in the United States?

$$\frac{A_t}{A_0} = e^{gt}$$

$A_t = working\ age\ fraction\ in\ 2030 = 0.54$
$A_0 = working\ age\ fraction\ in\ 2010 = 0.60$
$t = 2030 - 2010 = 20\ (years)$
$g = annual\ growth\ rate$

Hence

$$g = \frac{\ln\left(\frac{A_t}{A_0}\right)}{t}$$

Substituting$A_t = 0.54, A_0 = 0.60, t = 20$

$$g = -0.00527 = -0.527\%\ annually$$

In cobb-douglas production function,

$$Y = AK^{\alpha}L^{1-\alpha}$$

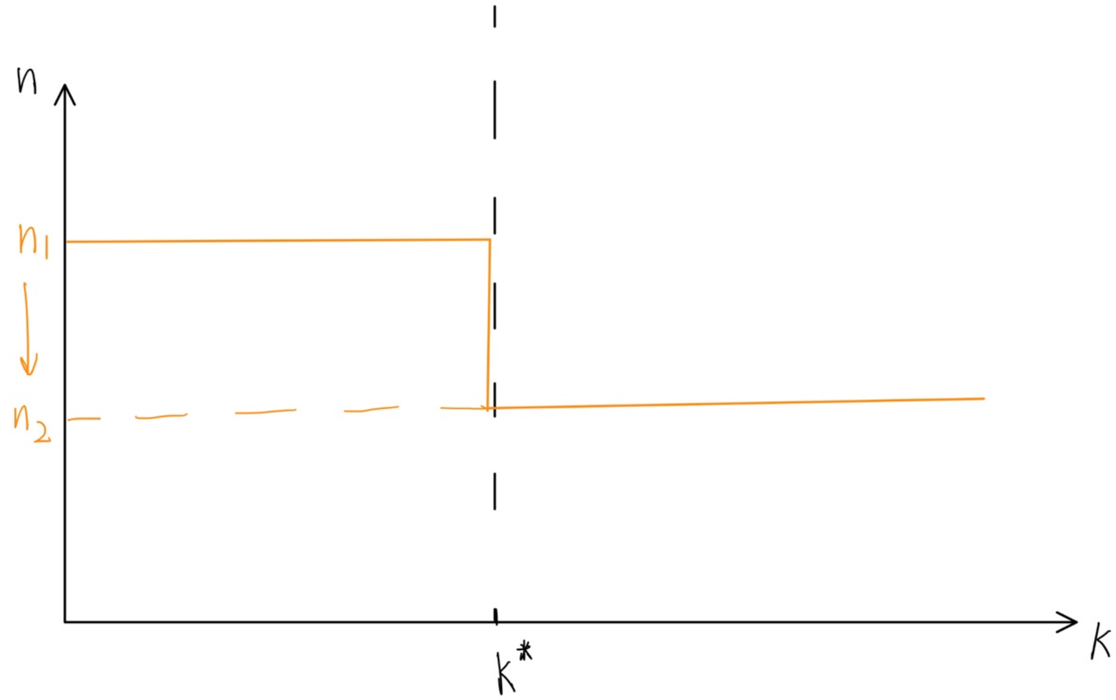
with $L_t = gL_{0,}\ A_t = A_0, K_t = K_0$

$$\Delta Y = Y_t - Y_0 = -0.527\%$$

GDP decrease 0.527 per year.

10 . Consider the Solow model as presented in the Weil textbook. Assume that the population can grow at two rates $\eta_1\ and\ \eta_2\ where\ \ \eta_1 > \eta_2$. Population growth depends on the level of output per capita and therefore on the level of capital per capita. Specifically, population growth grows at a rate η_1 when k is < $\bar{k}$ and slows down at a rate $\eta_2\ when\ k \geq\ \bar{k}$. Draw a diagram for this model. Assume that $(k +\ \delta)\bar{k}$ > $\gamma f(\bar{k})$ and that $(\eta_2$+$\delta)\bar{k} < \gamma f\left(\bar{k}\right)$. Explain what the model says about the steady state of the model. How relevant is this example to

the case of L&MICs?



At$(k + \delta)\bar{k} > \gamma f(\bar{k})$, the economy cannot maintain a steady state that maintains capital growth. which means that investment driven by output is insufficient and failed to offset capital depreciation and the needs of the population growth, causing capital per capita to remain low.

At $(\eta_2 + \delta)\bar{k} < \gamma f(\bar{k})$, economy exceed the threshold, and the population growth rate slows down. This slows the demand for new capital, allowing for a sustainable steady state where both capital and output grow at a balanced rate.

In relation with L&MIC, their typical low capital accumulation and high population growth often lead to poverty traps, preventing capital per capita from rising. In these countries, capital per capita may be below a certain threshold, resulting in rapid population growth and insufficient investment. To jump to a higher steady state, a suggestion is to increase capital accumulation through investments in infrastructure, education, and technology. This will allow them to slow population growth and increase capital per capita, eventually reaching the threshold that leads to sustainable growth.